

from every state

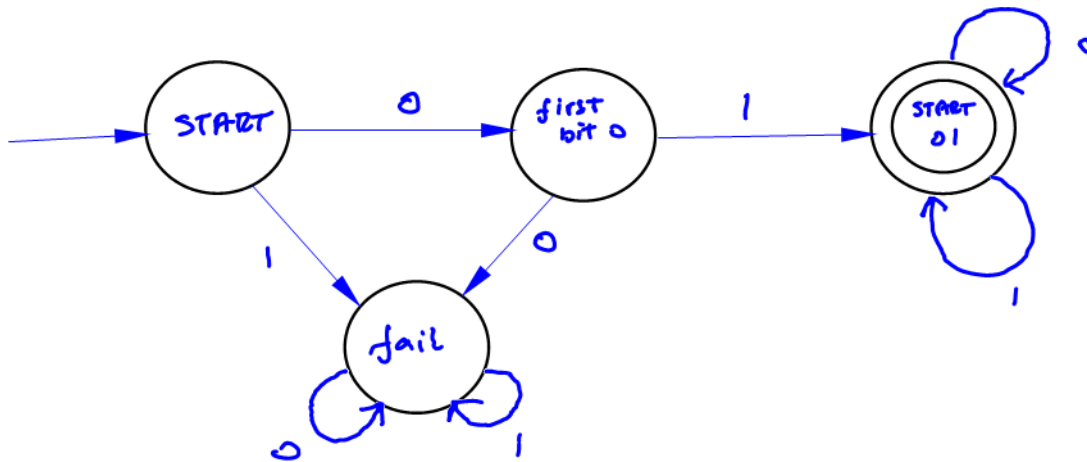


COMP 2121 DISCRETE MATHEMATICS

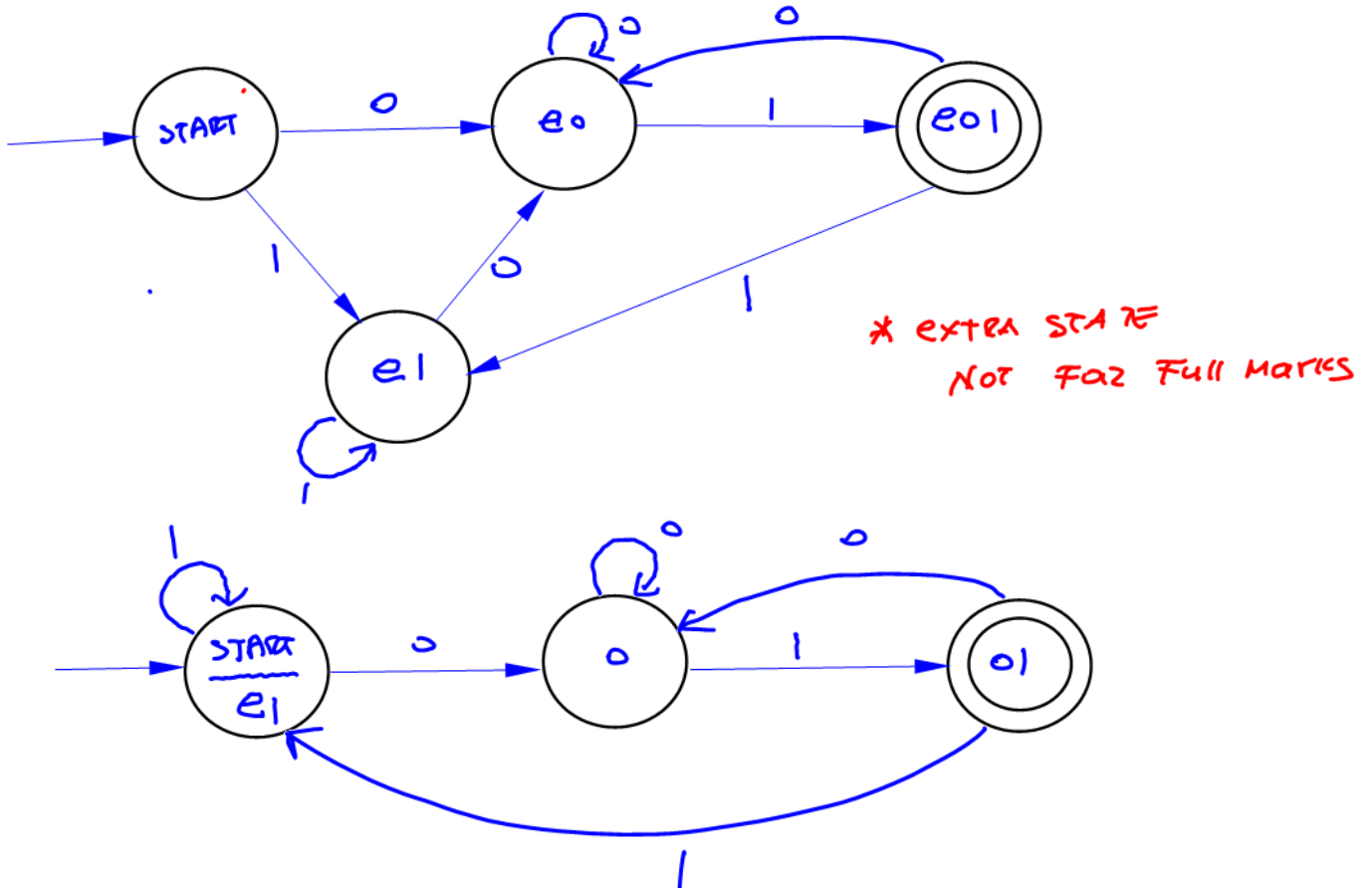
LAB 6

1. Design FSM with the given input alphabet that accepts the given set of strings:

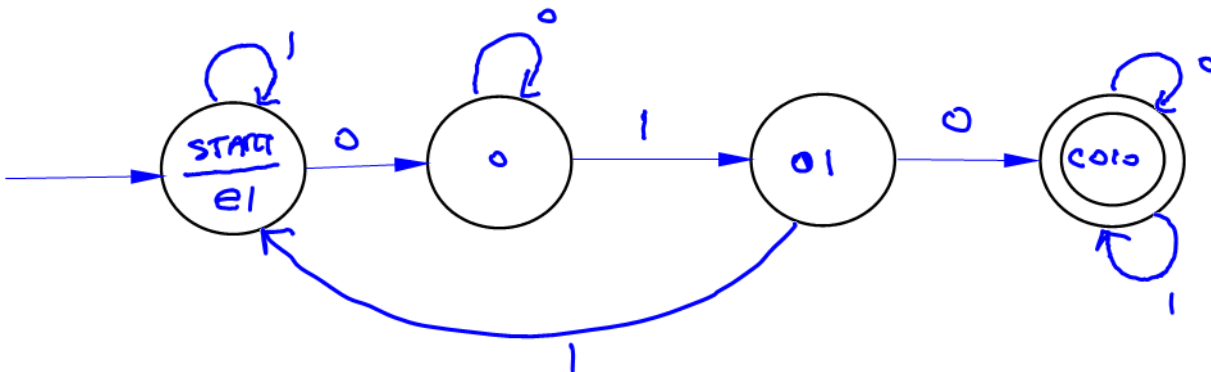
a) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that start with 01.



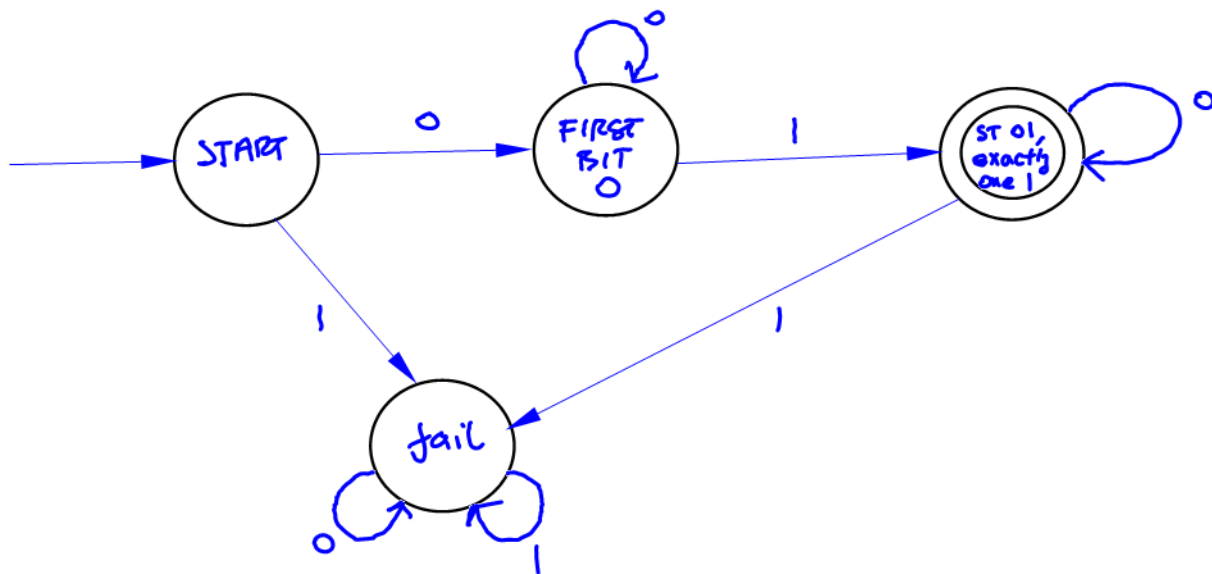
b) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that end with 01.



c) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that contain the pattern 010.

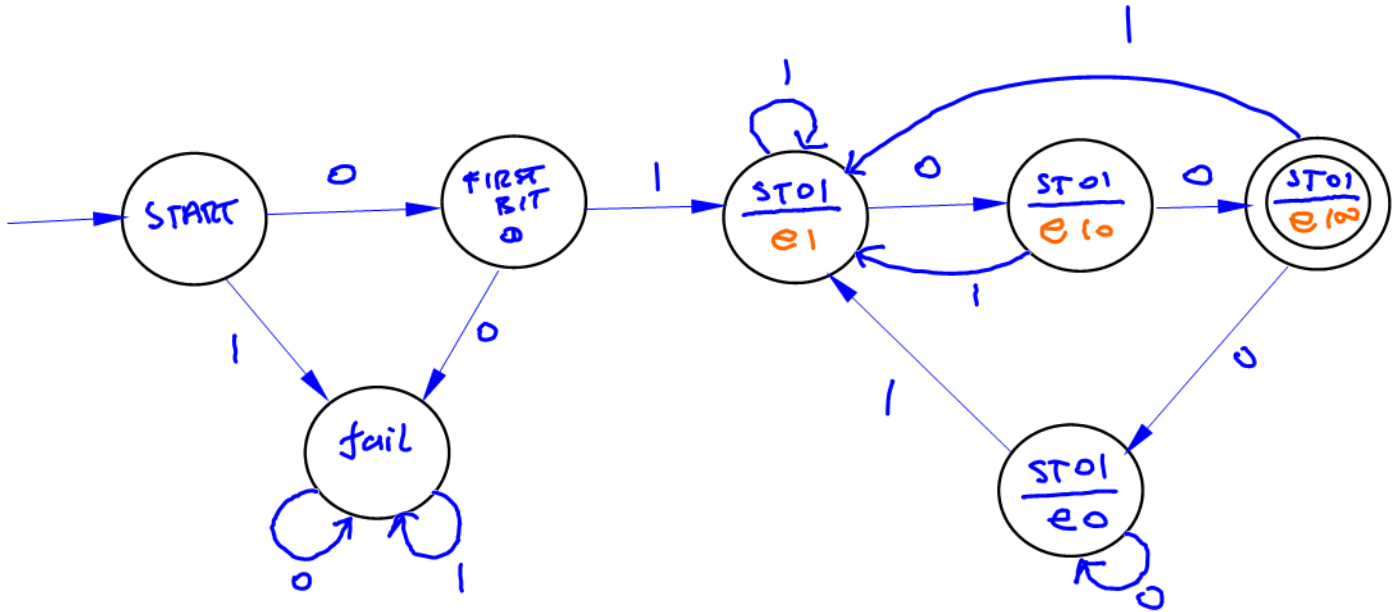


d) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that start with 0 and contain exactly one 1.



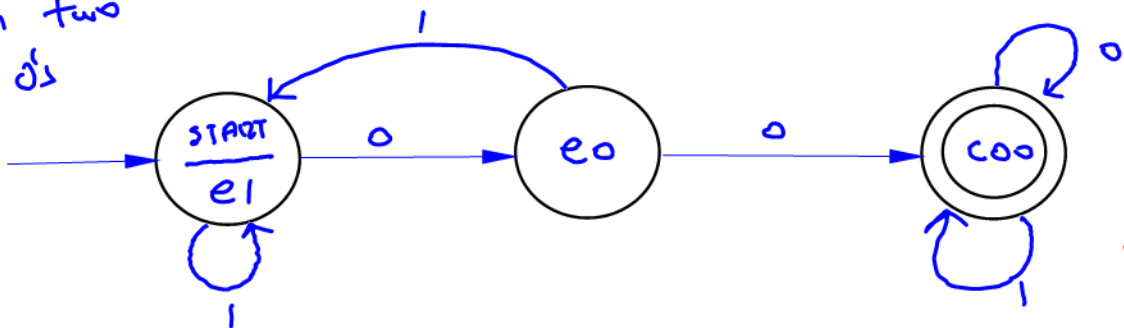
0 1 0 0

- e) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that start with 01 and end with 100.

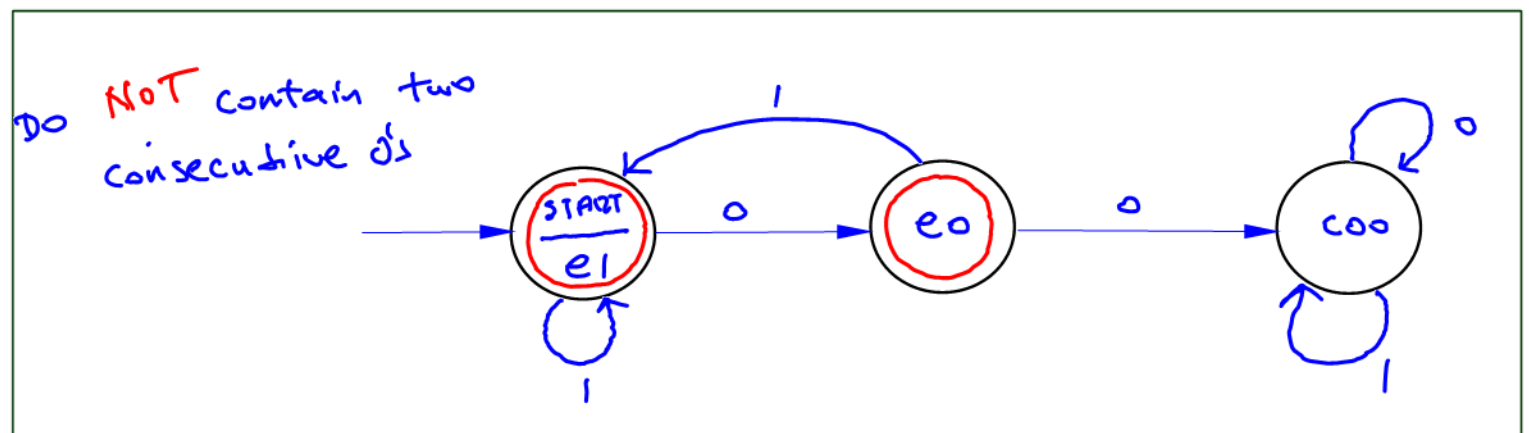


- f) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that do not contain two consecutive 0s.

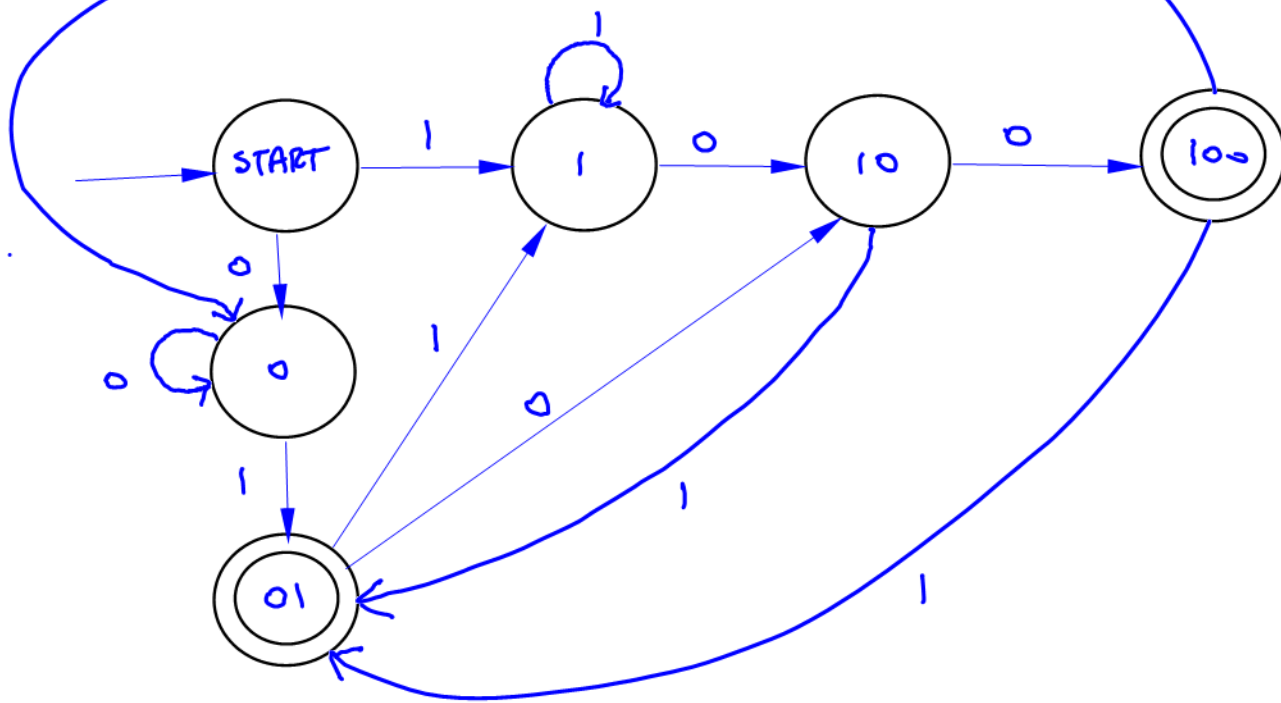
Do contain two consecutive 0s



to negate flip the states



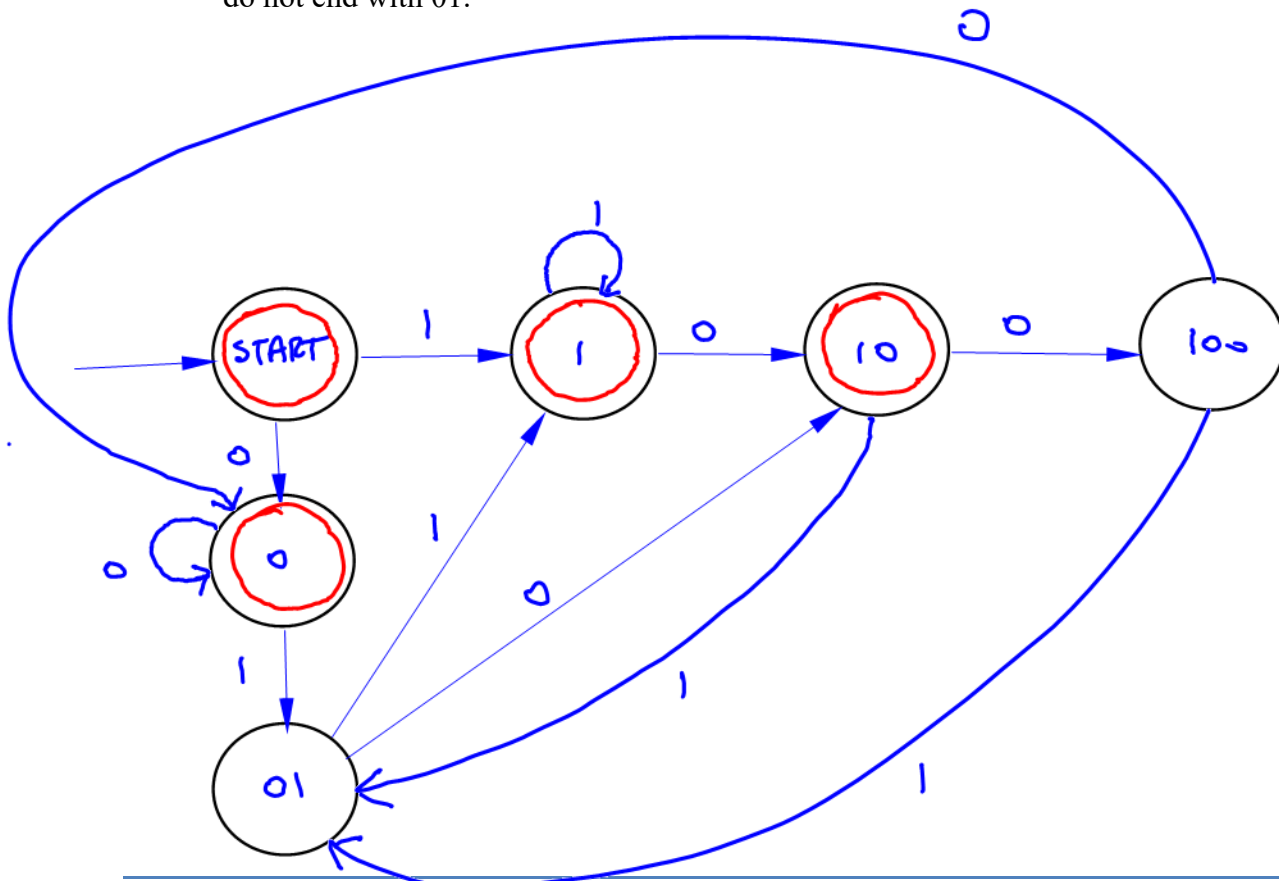
g) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that end with 100 or 01.



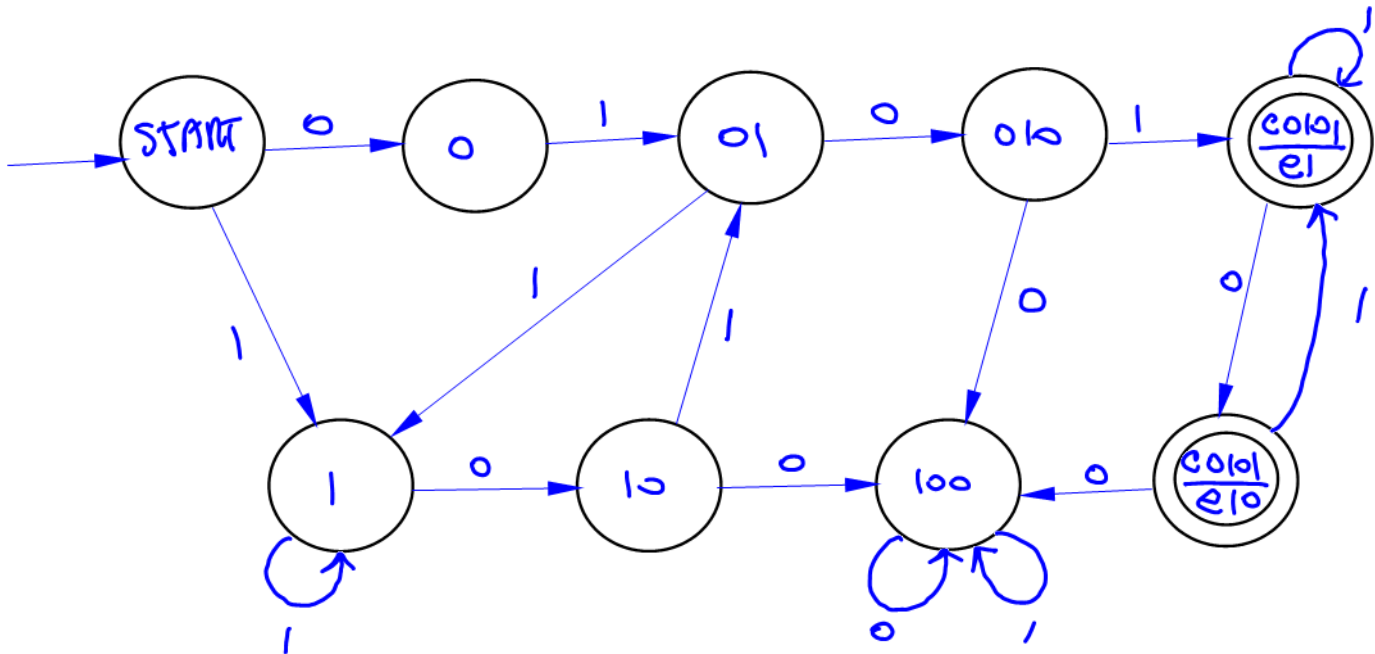
$E100 \text{ OR } E01$

NOT $E100$ AND NOT $E01 = \text{NOT}(E100 \text{ OR } E01)$

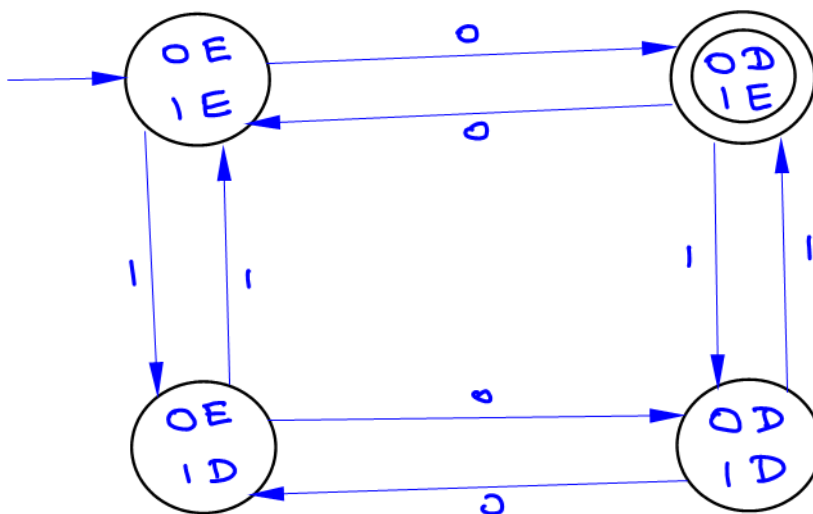
h) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that do not end with 100 and do not end with 01.



- i) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that contain 0101 but not 100.

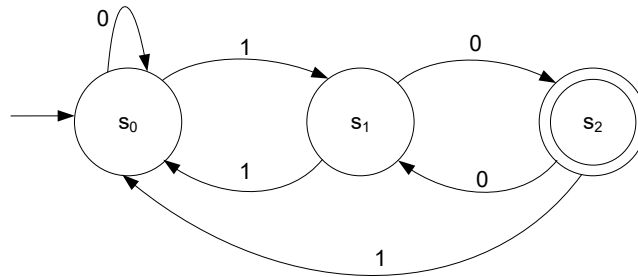


- j) Input alphabet = $\{0, 1\}$; Accepts the set of all strings containing an even number of 1s and an odd number of 0s.



EVEN - E
ODD - D

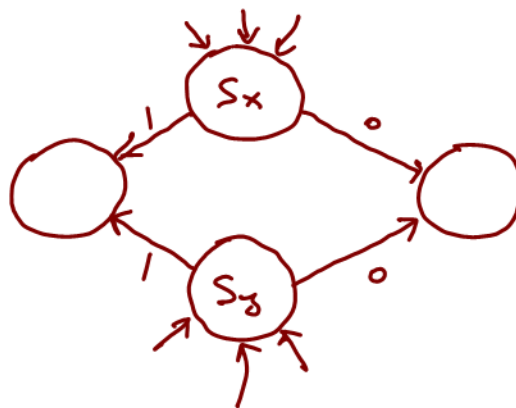
2. Characterize the set of strings accepted by the following machine:



contains an odd # of 1's and ends with 1 followed
by an odd # of 0's



NOTE, in this scenario s_x & s_z are the same state



If all of the problems can't be done in the lab period, students can work on them on their own.